Krishna Karthikeyan | BE23B033 | IIT Madras | KTH Stockholm, Sweden

EDUCATION AND SCHOLASTIC ACHIEVEMENTS

Program	Institute	% / CGPA	Year
Bachelor of Technology in Biological Engineering	Indian Institute of Technology, Madras	8.90	2027
Senior Secondary Class XII (CBSE)	Chettinad Vidyashram	94.8%	2023
Higher Secondary Class X (CBSE)	Chettinad Vidyashram	92%	2021

- Secured All India Rank **6025** among **150,000** aspirants in **JEE (Advanced)** conducted by IITs
- Secured **99.2%ile** in JEE Main examination among 1.1 million aspirants
- Awarded the prestigious KVPY Fellowship by DST, Govt of India with **All India Rank 48**
- Scored **1560 in the SAT**
- Scored **524/720** in NEET exam for entrance into MBBS programs
- Foundation in **Data Science and Applications** (online degree), with a **CGPA of 9.33**
- CBSE AISSE 10th Grade School Topper in French Language - 99/100
- AISSCE Topper 12th Grade National Topper in Chemistry - 100/100
- Participated in the exchange program in **KTH Stockholm, Sweden**

RELEVANT COURSES & SKILLS

Relevant Courses	Medical Devices, Medical Visualisation, Biomechanics of Human Movement, Clinical Applications of Biotech
Skills	Problem Solving, Analytical thinking, Attention to detail, Technical communication,
Software	Fusion360, Python, C, SolidWorks, MATLAB, OpenSim, VTK

PROFESSIONAL EXPERIENCE

Product Design Intern
vTitan
(May '25 - Jun '25)

- Gained hands-on knowledge of **medical device manufacturing processes**, including DFMA (Design for Manufacturing and Assembly), GD&T, and regulatory considerations.
- **Developed and refined 3D CAD models** of medical device components using Fusion360 and Solidworks software, with attention to dimensional accuracy and tolerances.
- Applied manufacturing and design principles to solve **real-world problem statements**, delivering functional, production-ready prototypes.

COURSE PROJECTS

EnviroFuel Research Project
(Apr '25 - Present)

- Conducted **hands-on experiments** to develop a bioethanol production process from waste paper through hydrolysis, incorporating innovative methods for saccharification and fermentation using **Bacillus sp. PM06**.
- Gained **Wet-Lab experience** and learned the usage of the following techniques/equipment: streaking, autoclaving, OD measurement, mechanical blending and working in LAF.

Medical Device Design
(Jun '25 – Oct '25)

- Designed a theoretical **multi-modal, non-invasive wearable stress monitoring system** integrating sweat-based cortisol sensing (microfluidics + MIP–OECT), electrodermal activity, HRV, SpO_2, temperature, and motion data for comprehensive physiological stress assessment.
- Demonstrated how **direct biochemical and neurophysiological biomarkers** outperform HRV-only approaches.

Medical Visualisation Program
(Oct '25 – Jan '26)

- Built a **Python-based medical imaging visualization tool** for neurosurgical planning using VTK/PyVista/PyQt5, supporting CT/CTA/MRA volume rendering, surface reconstruction, and interactive 3D views.
- Visualized **segmented cerebral vasculature** with anatomical labeling, artifact-aware preprocessing, and exportable clinical views (screenshots, fly-throughs)

Biomechanical Device Design
(Oct '25 – Jan '26)

- Conducted **comprehensive musculoskeletal gait analysis** (walking, jogging, crouch) using OpenSim and MATLAB, including inverse kinematics/dynamics, residual reduction, and static optimization to quantify joint moments, powers, and muscle activations
- Designed and evaluated a **passive ankle exoskeleton with a uniarticular spring**, optimizing resting length, stiffness, and moment arm to reduce plantarflexor activation and metabolic cost, while assessing comfort, pressure distribution, and real-world wearability

TEACHING EXPERIENCE

Workshop Organiser/ Facilitator
Jan '25

- Conducted a **1-day workshop** on "Biotech Frontiers: Discovery, Design, Destiny… ",for an academically diverse set of students, showcasing a strong commitment to knowledge dissemination.
- Engaged **30 participants** with **interactive lectures** and **hands-on lab activities**, including **gel electrophoresis**, ensuring practical understanding of recent and modern fascinating biotech concepts.

EXTRA-CURRICULARS

National Cadet Corps
B - Certificate
(Aug '23 - Mar '25)

- Completed a 10-day NCC camp, building discipline, physical fitness, and leadership skills through rigorous training and teamwork.
- Demonstrated a commitment to national service by upholding the values of responsibility, resilience, and dedication throughout the program.

IITM TV **Writing Head**	● Wrote and directed four short films, demonstrating creative storytelling, visual composition, and narrative design across diverse genres. ● Produced a short film that garnered 5.1K views on YouTube; currently serving as Chief Director, managing and mentoring a 10+ member filmmaking team.